

Tutorial 03

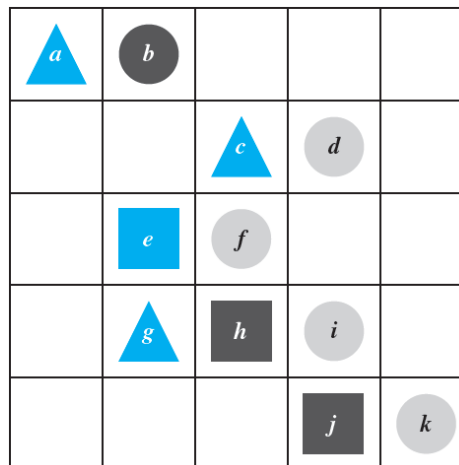
Predicates and Quantified Statements

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Tarski's World

- Tarski's World is a computer program developed by information scientists Jon Barwise and John Etchemendy to help teach the principles of logic.
- The program for Tarski's World provides pictures of blocks of various sizes, shapes, and colors, which are located on a grid.
- Shown in Figure 3.1.1 is a picture of an arrangement of objects in a two-dimensional Tarski world.

•Figure 3.1.1



Example 3.1.13 – *Investigating Tarski's World* continued

- The configuration can be described using logical operators and—for the two-dimensional version—notation such as:
 - Triangle(x), meaning “ x is a triangle,”
 - Blue(y), meaning “ y is blue,”
 - RightOf(x, y), meaning “ x is to the right of y (but possibly in a different row).”
- Individual objects can be given names such as a , b , or c .
- Determine the truth or falsity of each of the following statements. The domain for all variables is the set of objects in the Tarski world shown in Figure 3.1.1.

Example 3.1.13 – *Investigating Tarski's World*

continued

- a. $\forall t, \text{Triangle}(t) \rightarrow \text{Blue}(t)$.
- b. $\forall x, \text{Blue}(x) \rightarrow \text{Triangle}(x)$.
- c. $\exists y$ such that $\text{Square}(y) \wedge \text{RightOf}(d, y)$.
- d. $\exists z$ such that $\text{Square}(z) \wedge \text{Gray}(z)$.

 <i>a</i>	 <i>b</i>			
		 <i>c</i>	 <i>d</i>	
	 <i>e</i>	 <i>f</i>		
	 <i>g</i>	 <i>h</i>	 <i>i</i>	
			 <i>j</i>	 <i>k</i>

Exercise 3.1 – Problem 29

Rewrite each statement without using quantifiers or variables.
Indicate which are true and which are false.

- 29.** Let the domain of x be the set of geometric figures in the plane, and let $\text{Square}(x)$ be “ x is a square” and $\text{Rect}(x)$ be “ x is a rectangle.”
- a.** $\exists x$ such that $\text{Rect}(x) \wedge \text{Square}(x)$
 - b.** $\exists x$ such that $\text{Rect}(x) \wedge \sim \text{Square}(x)$
 - c.** $\forall x, \text{Square}(x) \rightarrow \text{Rect}(x)$

Exercise 3.1 – Problem 30

Rewrite each statement without using quantifiers or variables. Indicate which are true and which are false.

30. Let the domain of x be $\mathbf{Z}$, the set of integers, and let $\text{Odd}(x)$ be “ x is odd,” $\text{Prime}(x)$ be “ x is prime,” and $\text{Square}(x)$ be “ x is a perfect square.” (An integer n is said to be a **perfect square** if, and only if, it equals the square of some integer. For example, 25 is a perfect square because $25 = 5^2$.)

- a. $\exists x$ such that $\text{Prime}(x) \wedge \sim \text{Odd}(x)$
- b. $\forall x, \text{Prime}(x) \rightarrow \sim \text{Square}(x)$
- c. $\exists x$ such that $\text{Odd}(x) \wedge \text{Square}(x)$

Exercise 3.2 – Problem 24

- 24.** Rewrite the statements in each pair in if-then form and indicate the logical relationship between them.
- a.** All the children in Tom's family are female.
All the females in Tom's family are children.
 - b.** All the integers that are greater than 5 and end in 1, 3, 7, or 9 are prime.
All the integers that are greater than 5 and are prime end in 1, 3, 7, or 9.

Exercise 3.2 – Problem 36

- 36.** If $P(x)$ is a predicate and the domain of x is the set of all real numbers, let R be “ $\forall x \in \mathbf{Z}, P(x)$,” let S be “ $\forall x \in \mathbf{Q}, P(x)$,” and let T be “ $\forall x \in \mathbf{R}, P(x)$.”
- a.** Find a definition for $P(x)$ (but do not use “ $x \in \mathbf{Z}$ ”) so that R is true and both S and T are false.
 - b.** Find a definition for $P(x)$ (but do not use “ $x \in \mathbf{Q}$ ”) so that both R and S are true and T is false.

Exercise 3.2 – Problem 39-42

rewrite each statement of 39–44 in if-then form.

- 39.** Earning a grade of C– in this course is a sufficient condition for it to count toward graduation.
- 40.** Being divisible by 8 is a sufficient condition for being divisible by 4.
- 41.** Being on time each day is a necessary condition for keeping this job.
- 42.** Passing a comprehensive exam is a necessary condition for obtaining a master's degree.

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A is a sufficient condition for B $A \rightarrow B$

A is a necessary condition for B $B \rightarrow A$

Similarly, we have:

A is not a sufficient condition for B $\neg(\forall x, A \rightarrow B) \equiv \exists x, A \wedge \neg B$

A is not a necessary condition for B $\neg(\forall x, B \rightarrow A) \equiv \exists x, B \wedge \neg A$

Exercise 3.2 – Problem 45-48

Use the facts that the negation of a $\forall$ statement is a $\exists$ statement and that the negation of an if-then statement is an and statement to rewrite each of the statements 45–48 without using the word necessary or sufficient.

- 45.** Being divisible by 8 is not a necessary condition for being divisible by 4.
- 46.** Having a large income is not a necessary condition for a person to be happy.
- 47.** Having a large income is not a sufficient condition for a person to be happy.
- 48.** Being a polynomial is not a sufficient condition for a function to have a real root.